Huawei Hebbal Edge Access QoS Congestion Event Resolution SOP

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8298

Description

This Standard Operating Procedure (SOP) addresses the resolution of QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation, on Huawei Access equipment at the Hebbal Edge site. The issue is characterized by the ALARM_8298 alarm code, indicating a minor severity event. The underlying cause is rooted in the interaction between the QoS policy and the physical interface characteristics. In Huawei's Access equipment, the QoS policy is enforced through the Hierarchical QoS (H-QoS) mechanism, which classifies and prioritizes traffic based on the DSCP value. However, when the traffic rate exceeds the interface bandwidth, microbursts can occur, leading to interface tail drops and QoS queue starvation. This can be exacerbated by configuration issues, such as incorrect shaping or policing settings, or physical limitations, such as insufficient interface bandwidth or oversubscription.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible via remote hands or on-site personnel.
- Check the status of the redundant link and confirm that it is operational.
- Review the QoS policy configuration to identify any potential issues or conflicts.

Potential Root Causes

Cause	Probability
Incorrect QoS policy configuration	High
Physical interface bandwidth limitations	Medium
Traffic shaping or policing configuration issues	Low

Diagnostic Commands & Expected Outputs

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Command: display qos policy, Expected Output: QoS policy configuration details, including class maps, policy maps, and service policies., Description: This command displays the QoS policy configuration, allowing the analyst to identify potential issues or conflicts.

Command: display interface statistics, Expected Output: Interface statistics, including packet count, byte count, and error count., Description: This command displays the interface statistics, enabling the analyst to detect microbursts or interface tail drops.

Command: display qos queue statistics, Expected Output: QoS queue statistics, including packet count, byte count, and drop count., Description: This command displays the QoS queue statistics, allowing
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the analyst to identify QoS queue starvation.
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Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: display qos policy, Description: Review the QoS policy configuration to identify any potential issues or conflicts.

Step 2: Step: Adjust QoS policy configuration, Command: qos policy-map class police , Description: Adjust the QoS policy configuration to ensure correct shaping or policing settings.

Step 3: Step: Verify interface configuration, Command: display interface configuration, Description: Review the interface configuration to ensure correct settings and avoid physical limitations.

Post-Check Verification

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Command: display qos policy, Expected Output: Corrected QoS policy configuration details., Description: Verify that the QoS policy configuration has been corrected.

Command: display interface statistics, Expected Output: Normal interface statistics., Description: Verify that the interface statistics have returned to normal.

Command: display qos queue statistics, Expected Output: Normal QoS queue statistics., Description: Verify that the QoS queue statistics have returned to normal.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo qos policy-map class police , Description: Restore the original QoS policy configuration if the fix fails.

- Step: Restore original interface configuration, Command: undo interface configuration, Description: Restore the original interface configuration if the fix fails.